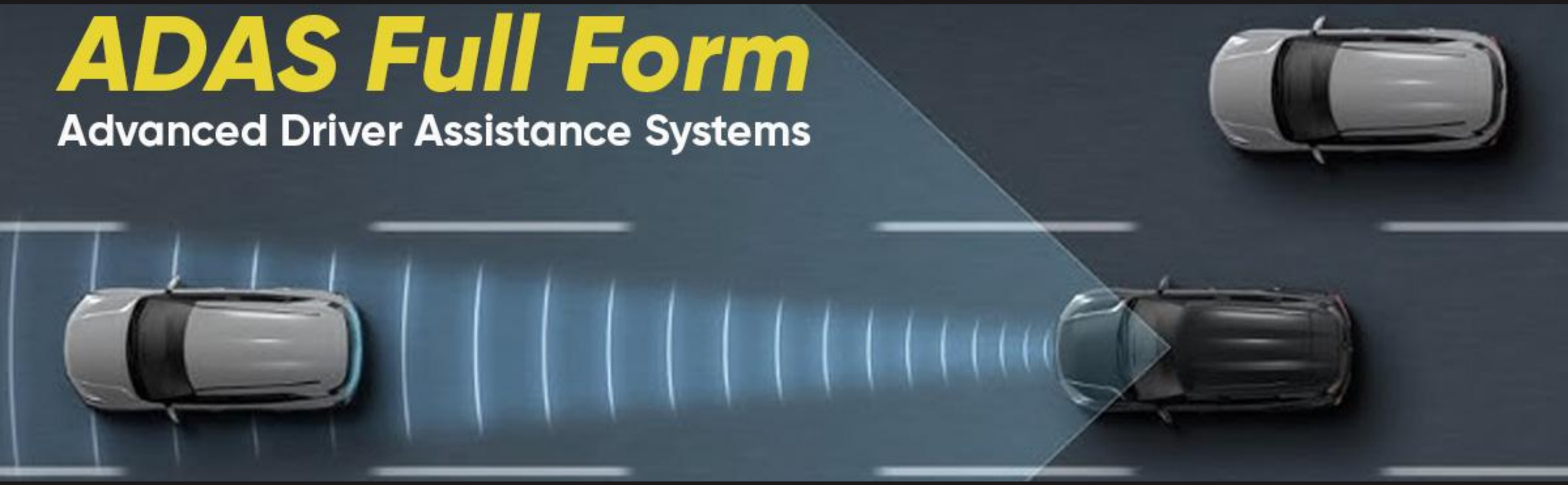
FOP
MINI PROJECT

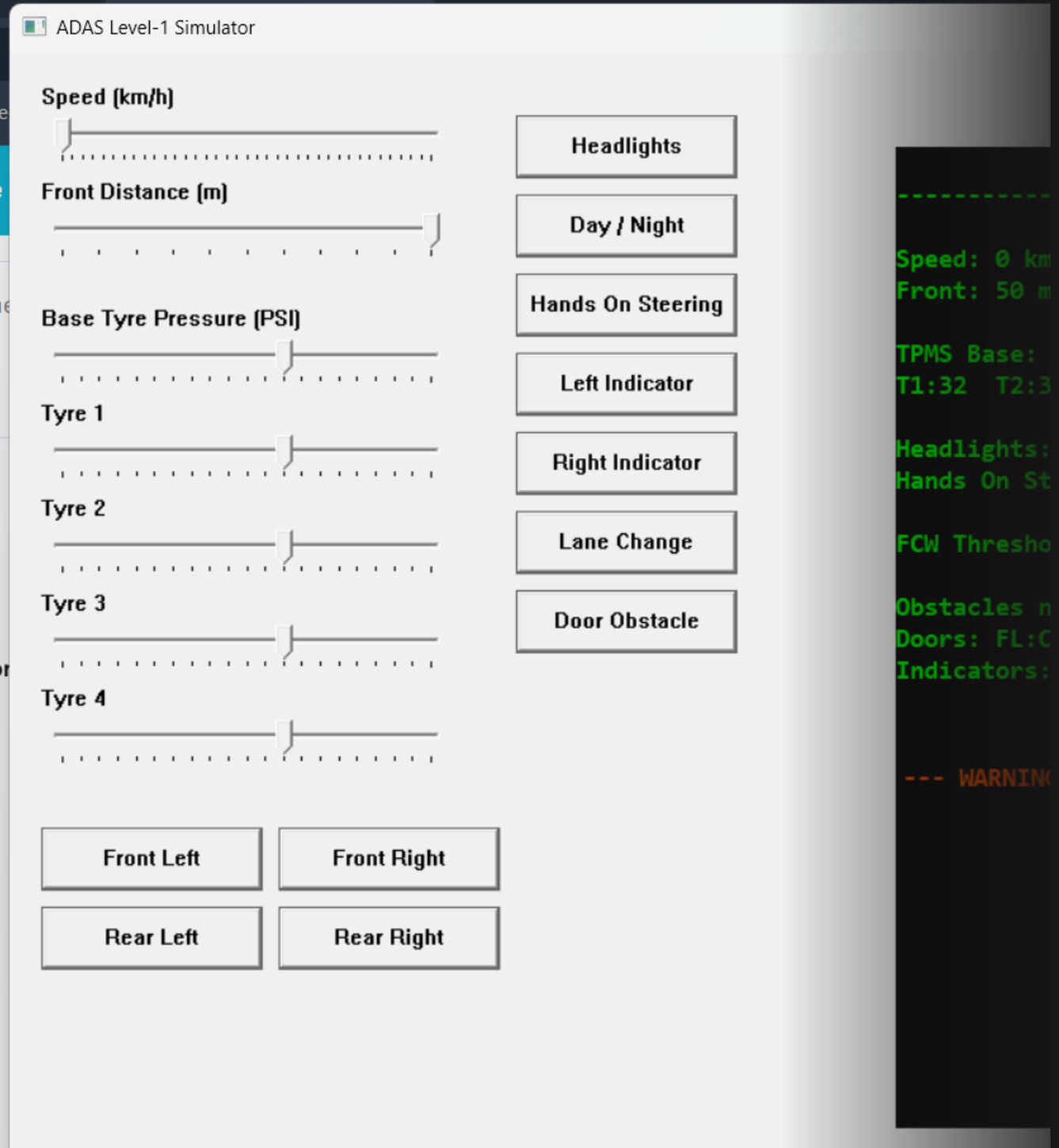
ADAS LEVEL - 1 (SIM)

This project presents a Level-1 Advanced Driver Assistance System (ADAS) simulation developed using C and the Win32 API. The application models essential vehicle safety and driver-assistance features through an interactive graphical interface.



ABOUT THE SIMULATION

The simulator allows real-time control of vehicle parameters such as speed, distance, tyre pressure, door states, indicators, and driver inputs, while dynamically evaluating safety conditions. Based on these inputs, the system generates visual warnings on a custom Multi-Information Display (MID) and audible alerts with different priority levels.



ADAS LEVEL-1 SIMULATION USING C & WIN32 API

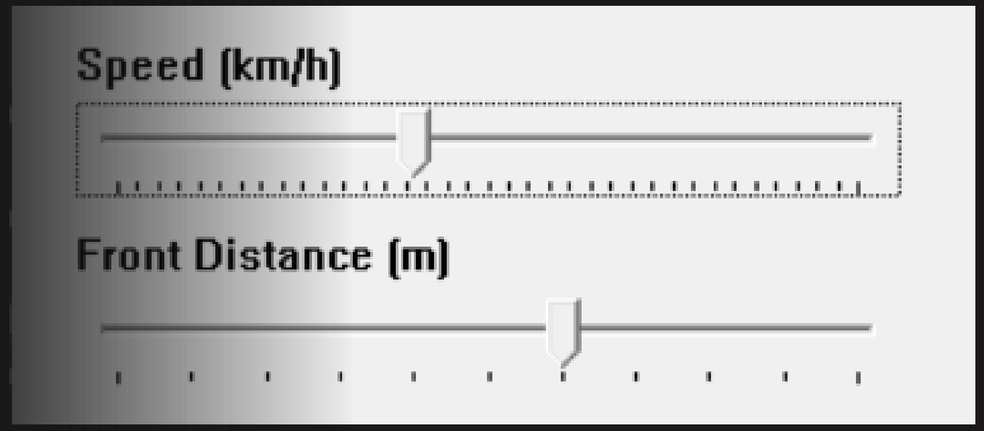
- Interactive ADAS simulator built in C
- Event-driven Win32 GUI
- Focus on safety logic, warnings, and driver interaction

GLOBAL STATE MANAGEMENT

VARIABLES TRACKING VEHICLE CONDITIONS

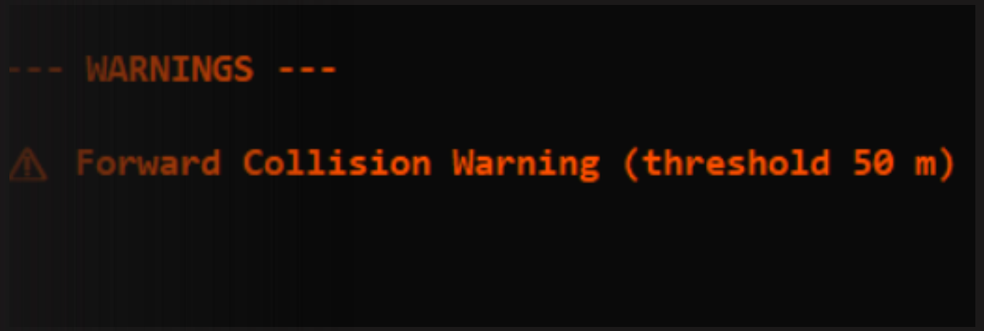
- Vehicle speed variable
- Front & rear distance values
- Base tyre pressure reference
- Individual tyre pressure array
- Boolean flags for:
  - Headlights
  - Night mode
  - Hands-on steering
  - Indicators
  - Door obstacles
- Ensures synchronized ADAS behavior across all features

FEATURES INCLUDED



SPEED & DISTANCE HANDLING

- Speed controlled using trackbar (slider)
- Front distance slider simulates obstacle proximity
- Values updated via WM_HSCROLL
- Real-time data reflected on MID display



FORWARD COLLISION WARNING (FCW)

- Speed compared with front distance
- Warning thresholds adapt dynamically
- Conditions:
- High speed + low distance → Critical warning
- Medium risk → Caution warning
- Triggers:
- MID warning message
- High-priority audible beep

FEATURES INCLUDED

```
--- WARNINGS ---

⚠ Low Tyre Pressure
```

TPMS (TYRE PRESSURE MONITORING SYSTEM)

- Base tyre pressure defined in code
- Four independent tyre pressure variables
- Continuous comparison with base pressure
- If pressure < threshold:
- TPMS warning displayed
- Medium-priority beep triggered

```
--- WARNINGS ---

⚠ Lane Change Warning: Use Indicator
```

LANE CHANGE ASSIST LOGIC

- Lane change request validated in code
- Condition check:
  - Indicator must be ON
- If indicator OFF:
  - Warning displayed on MID
  - Audible alert generated
- Simulates real ADAS driver-intent validation

FEATURES INCLUDED

```
--- WARNINGS ---

⚠ Door opening blocked: obstacle or vehicle moving
```

DOOR SAFETY SYSTEM

- Individual door open/close buttons
- Safety checks in code:
  - Vehicle speed must be zero
  - No obstacle detected
- If unsafe:
  - Door action blocked
  - Warning shown on MID
  - Beep alert triggered

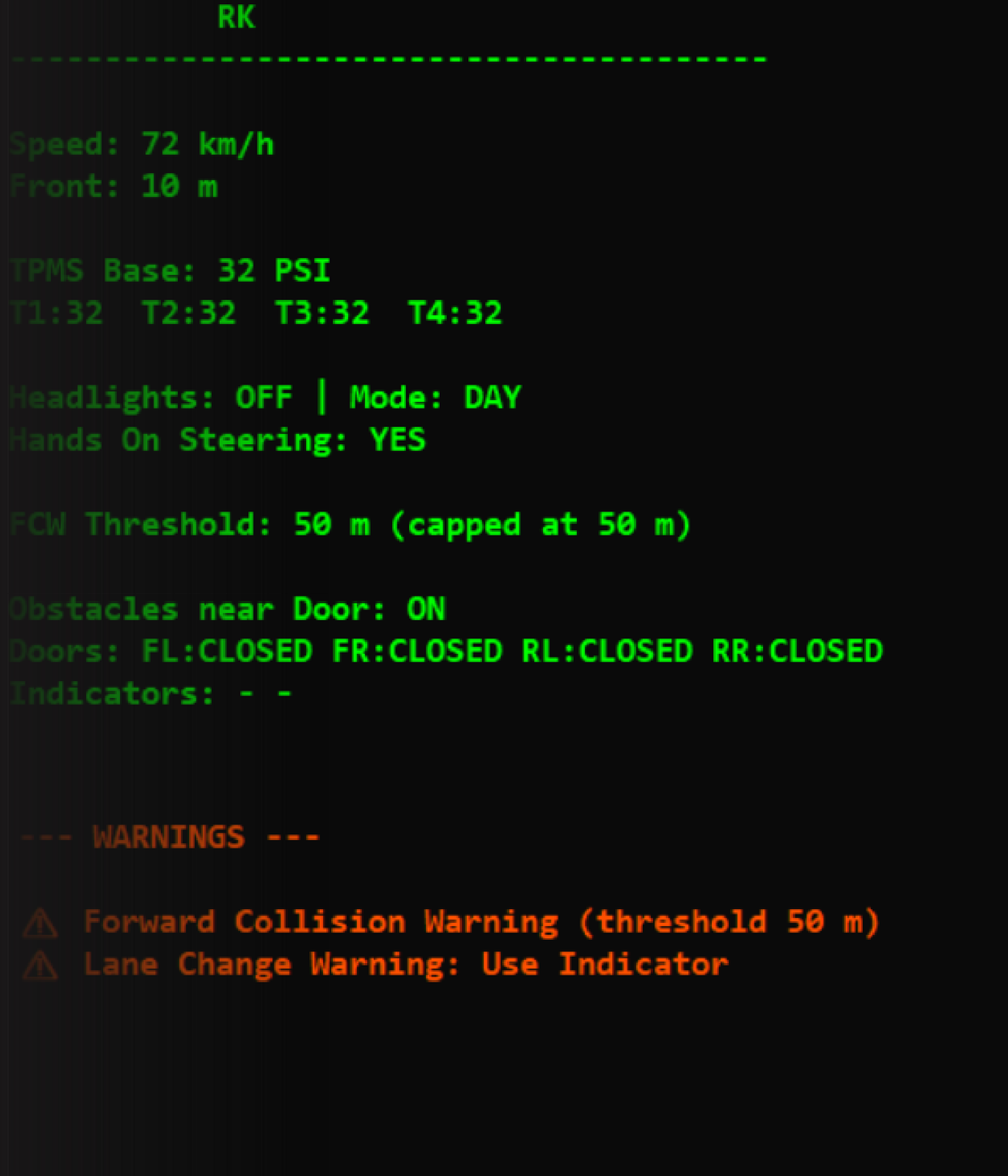
```
--- WARNINGS ---

⚠ Headlights ON (day)
```

HEADLIGHTS ASSIST

- Lane change request validated in code
- Condition check:
  - Indicator must be ON
- If indicator OFF:
  - Warning displayed on MID
  - Audible alert generated
- Simulates real ADAS driver-intent validation

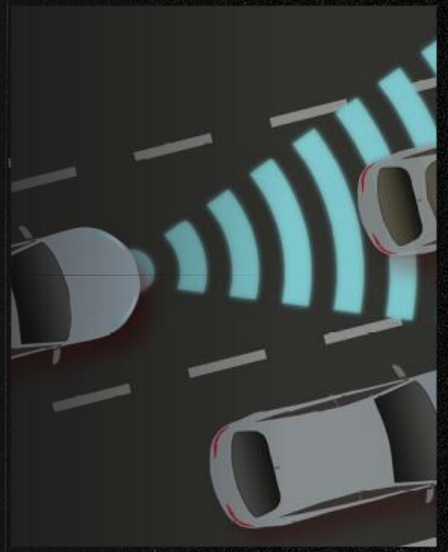
MID (MULTI-INFORMATION DISPLAY)

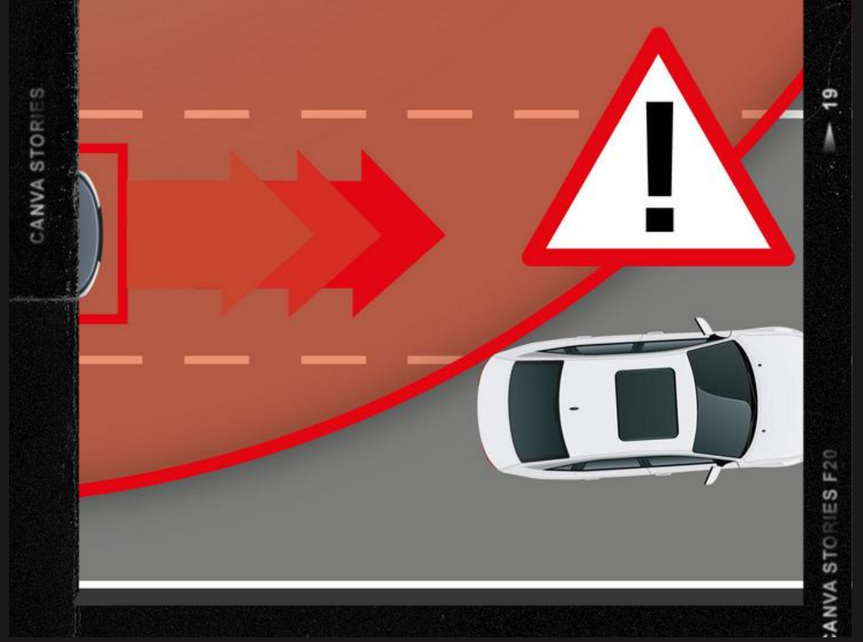
- Custom drawn using TextOut / DrawText
- Displays:
  - Speed
  - Distance
  - TPMS status
  - Door states
  - ADAS warnings
  - Headlight Assist
- Centralized information display logic
- Prevents clutter using priority-based messages



WARNING PRIORITY SYSTEM

- High Priority:
  - Collision warning
  - Unsafe door operation
- Medium Priority:
  - TPMS alerts
- Low Priority:
  - Informational messages
- Each priority has a distinct beep pattern
- Uses Win32 Beep() function
- Frequency & duration vary per warning type



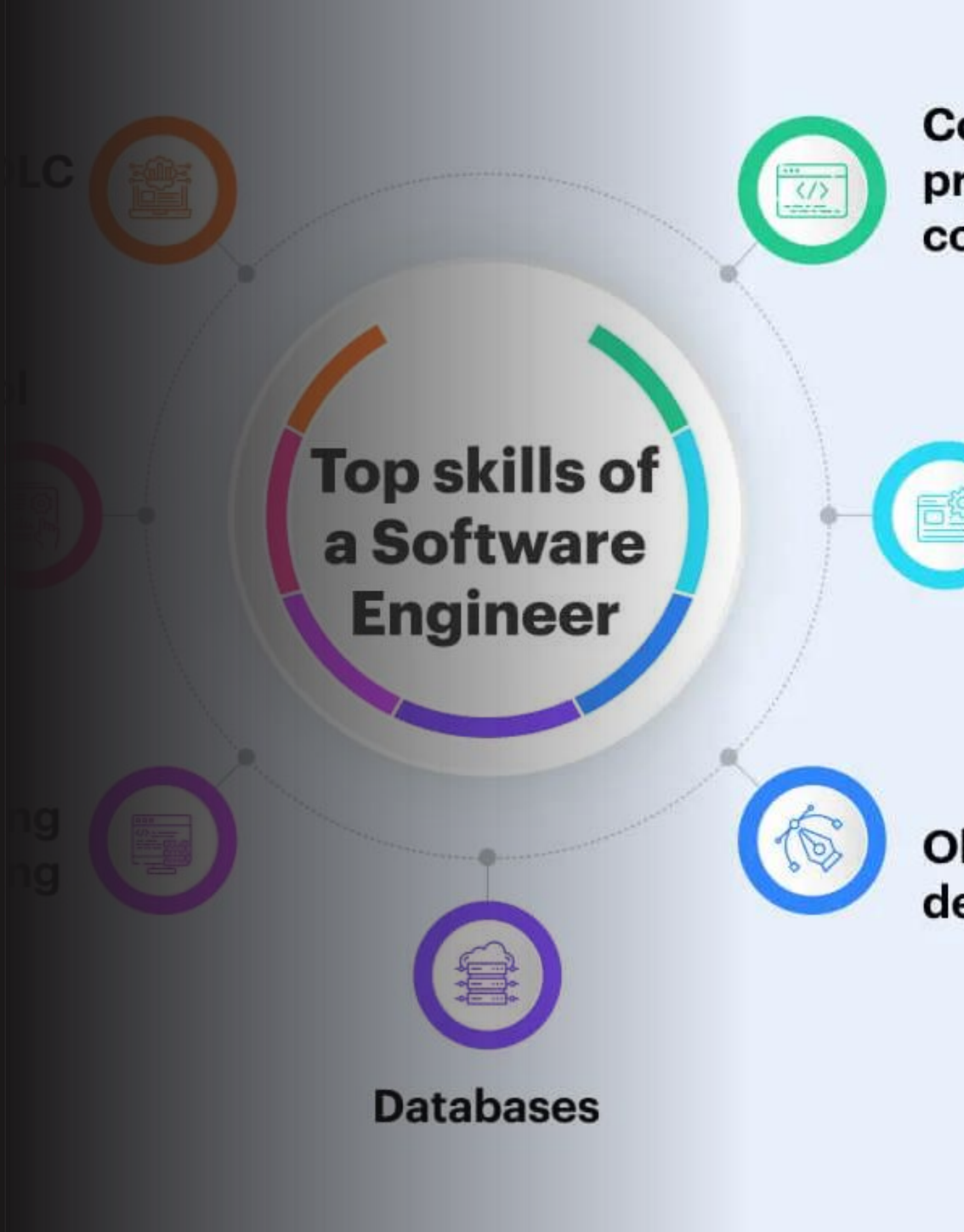


SAFETY-FIRST DESIGN

- No door opening at non-zero speed
- Lane change restricted without indicator
- Collision warnings adapt to speed
- Continuous monitoring of all vehicle parameters

LEARNING OUTCOMES

- Win32 GUI programming in C
- Real-time state handling
- ADAS logic simulation
- Automotive safety system modeling
- Event-based software design



Team Members & their contributions

- Rahul Krishna (1262250870) – Code & Presentation
- Akshat Yadav (1262253358) - Code
- Kuhu Shrivastava (1262253764) - Poster
- Vaishnavi Sapkal (1262251869) - Poster
- Rushil Tolia (1262252066) - Presentation

THANK YOU